

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	2024 / Slot C	Date:	08-08-2026

Code of Conduct

I A Roja [192424226] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A Roja

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. **Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} | \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.**

Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} | \text{data})$.

The formula for Maximum Likelihood Estimate of a bigram probability is:

$$P(\text{science} \mid \text{data}) = C(\text{data science}) / C(\text{data})$$

Given:

$$C(\text{data science}) = 3$$

$$C(\text{data}) = 3$$

Therefore,

$$P(\text{science} \mid \text{data}) = 3 / 3$$

$$= 1.0$$

Hence, $P(\text{science} \mid \text{data}) = 1.0$ or 100%.

Interpretation:

The probability of predicting "science" after the word "data" is 1.0. This means that in the given training corpus, every occurrence of "data" is followed by "science". Therefore, the mobile keyboard can confidently predict "science" whenever the user types "data". A higher probability indicates greater confidence in the next-word prediction. However, in a real-world keyboard, relying only on MLE may fail when an unseen word sequence occurs.

- 2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.**

Apply the Backoff Model for the unseen sequence "data science improves".

The sequence is:

data → science → improves

The required trigram probability is:

$$P(\text{improves} \mid \text{data science})$$

The trigram "data science improves" does not occur in the given corpus. Therefore, its trigram probability is zero.

The model then backs off to the lower-order bigram probability:

$$P(\text{improves} \mid \text{science})$$

The given corpus contains:

science is

science drives

but it does not contain:

science improves

Therefore, the bigram probability is also zero.

The model then backs off to the unigram probability:

$P(\text{improves})$

The word "improves" does not occur in the given corpus. Therefore:

$P(\text{improves}) = 0$

Hence, the estimated probability is:

$P(\text{improves} \mid \text{data science}) \approx 0$

Interpretation:

Backoff is important in real-time predictive text systems because users frequently type new or unseen combinations of words. When a trigram is unavailable, the model can use bigram information, and if that is also unavailable, it can use unigram information. This prevents the prediction system from completely failing when it encounters unseen sequences. In a practical system, smoothing techniques can also be combined with backoff to assign small non-zero probabilities to unseen words.

3. Using Deleted Interpolation, calculate the probability of the sequence "data science is".

Discuss how interpolation balances reliability and coverage in sparse datasets.

Using Deleted Interpolation, calculate the probability of the sequence "data science is".

Deleted interpolation combines trigram, bigram, and unigram probabilities using the given interpolation weights.

The formula is:

$P(\text{is} \mid \text{data science}) = \lambda_1 P_3(\text{is} \mid \text{data science})$

- $\lambda_2 P_2(\text{is} \mid \text{science})$
- $\lambda_3 P_1(\text{is})$

Given:

$$\lambda_1 = 0.5$$

$$\lambda_2 = 0.3$$

$$\lambda_3 = 0.2$$

The given prediction probability after "science" is:

$$P(\text{is}) = 0.66$$

Since separate numerical trigram, bigram, and unigram probabilities for "is" are not provided, the supplied probability 0.66 can be used as the available estimate for the three components.

Therefore:

$$P(\text{is} \mid \text{data science}) = 0.5(0.66) + 0.3(0.66) + 0.2(0.66)$$

$$= 0.33 + 0.198 + 0.132$$

$$= 0.66$$

Therefore, the estimated probability is:

$$P(\text{is} \mid \text{data science}) = 0.66$$

Interpretation:

Deleted interpolation combines information from trigram, bigram, and unigram models. The trigram provides detailed contextual information, while the bigram and unigram models provide additional coverage when higher-order n-grams are sparse. Therefore, interpolation provides a balance between reliability and coverage, making predictions more robust for sparse datasets.

- 4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.**

Calculate the Entropy of the distribution $P(\text{is}) = 0.66$ and $P(\text{drives}) = 0.33$.

Entropy measures the uncertainty in a probability distribution.

The formula is:

$$H = -\sum P(x) \log_2 P(x)$$

Therefore:

$$H = -[0.66 \log_2(0.66) + 0.33 \log_2(0.33)]$$

Using approximate values:

$$\log_2(0.66) \approx -0.599$$

$$\log_2(0.33) \approx -1.599$$

Therefore:

$$H = -[(0.66 \times -0.599) + (0.33 \times -1.599)]$$

$$H \approx -[-0.395 - 0.528]$$

$$H \approx 0.923 \text{ bits}$$

Therefore, the entropy is approximately:

$$H \approx 0.92 \text{ bits}$$

Note: The given probabilities add up to 0.99 instead of 1.00 because they are rounded values. Therefore, the calculated entropy is an approximate value.

Interpretation:

The entropy of approximately 0.92 bits indicates a moderate level of uncertainty between the possible next words "is" and "drives". Since "is" has a higher probability than "drives", the keyboard has more confidence in predicting "is". Lower entropy indicates lower uncertainty and higher prediction confidence, while higher entropy indicates greater uncertainty. Thus, entropy is useful for evaluating how confident an intelligent keyboard is when selecting the next-word prediction.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. **Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.**

Given Sentence

"Book a flight ticket now."

POS tags:

Book/VB

a/DT

flight/NN

ticket/NN

now/RB

Explanation:

Book is tagged as VB (Verb) because it is used as an action or command. The sentence is an imperative sentence in which the user is asking the chatbot to book a flight.

a is tagged as DT (Determiner) because it specifies the noun "flight".

flight is tagged as NN (Noun) because it represents a thing or concept.

ticket is tagged as NN (Noun) because it is a noun and represents the object being booked.

now is tagged as RB (Adverb) because it indicates the time of the action.

Sentence 2:

"This book is interesting."

POS tags:

This/DT

book/NN

is/VBZ

interesting/JJ

Explanation:

This is tagged as DT (Determiner) because it specifies the noun "book".

book is tagged as NN (Noun) because it refers to an object and acts as the subject of the sentence.

is is tagged as VBZ because it is the third-person singular present form of the verb "be".

interesting is tagged as JJ (Adjective) because it describes the noun "book".

The word "book" has two different POS tags depending on its context:

"Book a flight" → Book/VB

"This book is interesting" → book/NN

Thus, contextual information is important for resolving lexical ambiguity.

2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.

Given:

$$P(\text{book} \mid \text{VB}) = 0.6$$

$$P(\text{book} \mid \text{NN}) = 0.4$$

$$P(\text{Start} \rightarrow \text{VB}) = 0.5$$

$$P(\text{Start} \rightarrow \text{NN}) = 0.5$$

For the Verb (VB) interpretation:

$$P(\text{VB}, \text{book}) = P(\text{Start} \rightarrow \text{VB}) \times P(\text{book} \mid \text{VB})$$

$$= 0.5 \times 0.6$$

$$= 0.30$$

Therefore:

$$\text{Likelihood of book being VB} = 0.30$$

For comparison, the Noun (NN) interpretation is:

$$P(\text{NN}, \text{book}) = P(\text{Start} \rightarrow \text{NN}) \times P(\text{book} \mid \text{NN})$$

$$= 0.5 \times 0.4$$

$$= 0.20$$

Therefore:

$$\text{Likelihood of book being NN} = 0.20$$

Comparison:

$$\text{VB} = 0.30$$

$$\text{NN} = 0.20$$

Since 0.30 is greater than 0.20, the HMM favors the Verb (VB) tag.

Explanation:

The model gives equal starting probabilities to VB and NN. However, the probability of observing "book" when the tag is VB is higher than when the tag is NN. Therefore, the HMM assigns a higher probability to the Verb interpretation. The grammatical context also supports this because "Book a flight ticket now" is an imperative sentence.

3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.

Rule-Based Tagging:

Rule-Based Tagging uses manually designed linguistic rules to assign POS tags. For example, a rule can state that a word appearing before a noun may function as a determiner. It can resolve ambiguity when appropriate rules have been created, but it requires many rules for handling complex language patterns.

Stochastic or HMM Tagging:

HMM tagging uses probabilities learned from a training corpus. It considers the probability of a word being associated with a particular POS tag and can use contextual information to select the most likely tag. For example, "book" can be identified as VB or NN depending on the surrounding words and learned probabilities.

Comparison:

Rule-Based Tagging depends mainly on manually created rules, while HMM tagging depends on statistical probabilities learned from data. Rule-based systems can be difficult to maintain when the language becomes complex. HMM systems are more adaptable and can handle ambiguity using probabilities

Recommendation:

For a large-scale customer support chatbot, Stochastic/HMM tagging is more suitable because it can learn patterns from large amounts of training data and handle ambiguous words more effectively. It can also be updated by retraining the model when new language patterns appear.

4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

Standardized POS tagsets such as the Penn Treebank POS Tagset provide a consistent way of representing grammatical categories such as nouns, verbs, adjectives, and adverbs. This helps the chatbot understand the grammatical structure of user messages.

POS tagging is useful for intent detection because it helps distinguish actions from objects. For example, in "Book a flight", "book" is a verb and indicates an action related to flight reservation. In "This book is interesting", "book" is a noun and refers to an object.

POS information also helps in response generation because the chatbot can identify important actions, entities, and relationships in a sentence. Correct POS tagging reduces ambiguity and improves language understanding.

Standardized word classes also make it easier to train NLP models because words are represented using consistent grammatical categories. This improves feature extraction, intent classification, response generation, and overall chatbot accuracy.

Therefore, POS tagsets and word classes play an important role in building reliable and efficient AI-powered customer support chatbots.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

- 1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.**

The given sentence is:

Economic growth increases employment.

Initial POS tags:

economic/JJ

growth/NN

increases/NNS

employment/NN

The given transformation rule is:

Change NNS to VBZ if the preceding word is tagged as NN.

The word "increases" is initially tagged as NNS. Its preceding word is "growth", which is tagged as NN. Therefore, the transformation rule is applied.

increases/NNS → increases/VBZ

Corrected POS tag sequence:

economic/JJ

growth/NN

increases/VBZ

employment/NN

The correction is linguistically appropriate because "increases" functions as the main verb of the sentence. The word "growth" is a singular noun acting as the subject, so the verb should have the third-person singular present-tense form "increases", which is represented by the POS tag VBZ.

2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.

The initial tagging is:

economic/JJ growth/NN increases/NNS employment/NN

The tag economic/JJ is correct because "economic" describes the noun "growth" and therefore functions as an adjective.

The tag growth/NN is correct because "growth" is a singular noun and acts as the subject of the sentence.

The tag increases/NNS is incorrect. NNS represents a plural noun, but "increases" is functioning as a verb in this sentence. It describes an action performed by the subject "growth".

The tag employment/NN is correct because "employment" is a noun and functions as the object of the verb "increases".

Therefore, the corrected sequence is:

economic/JJ growth/NN increases/VBZ employment/NN

The sentence structure is:

Subject + Verb + Object

growth + increases + employment

Since "growth" is singular, the verb "increases" agrees with the subject and is correctly tagged as VBZ.

3. Assuming the corpus contains the words:

- **economic (120 occurrences)**
- **growth (450 occurrences)**
- **increases (210 occurrences)**
- **employment (380 occurrences)**

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging

Given corpus frequencies:

economic = 120

growth = 450

increases = 210

employment = 380

First, calculate the total frequency:

$$\text{Total} = 120 + 450 + 210 + 380$$

$$\text{Total} = 1160$$

The probability of each word can be calculated as:

$$P(\text{word}) = \text{Frequency of word} / \text{Total frequency}$$

For economic:

$$P(\text{economic}) = 120 / 1160$$

$$= 0.1034$$

For growth:

$$P(\text{growth}) = 450 / 1160$$

$$= 0.3879$$

For increases:

$$P(\text{increases}) = 210 / 1160$$

$$= 0.1810$$

For employment:

$$P(\text{employment}) = 380 / 1160$$

$$= 0.3276$$

Word frequency distribution:

$$\text{economic} = 120 \text{ occurrences} = 0.1034$$

$$\text{growth} = 450 \text{ occurrences} = 0.3879$$

$$\text{increases} = 210 \text{ occurrences} = 0.1810$$

$$\text{employment} = 380 \text{ occurrences} = 0.3276$$

Therefore, the highest-frequency word is "growth", with 450 occurrences, while the lowest-frequency word is "economic", with 120 occurrences.

Frequency information helps probabilistic POS tagging because frequently observed words and their contexts provide stronger statistical evidence for predicting their grammatical categories. For example, if "increases" frequently appears after singular nouns and behaves as a verb, the POS tagger can assign VBZ with higher confidence.

However, word frequency alone cannot determine the POS tag because a word may have multiple grammatical roles depending on its context. Therefore, probabilistic taggers use both word frequencies and contextual information.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

Transformation-Based Tagging improves the initial POS tagging by applying learned correction rules based on the context of a word.

Initially, the system produces:

economic/JJ

growth/NN

increases/NNS

employment/NN

The tagger identifies that "increases/NNS" is inconsistent with the surrounding grammatical structure. The learned rule checks whether the preceding word is tagged as NN. Since "growth" is tagged as NN, the rule changes:

increases/NNS → increases/VBZ

The final sequence becomes:

economic/JJ growth/NN increases/VBZ employment/NN

This improves the tagging accuracy because the transformation rule corrects a systematic tagging error using contextual information.

Entropy can be used to measure the uncertainty of POS tag assignments. The entropy formula is:

$$H = -\sum P(t) \log_2 P(t)$$

where $P(t)$ represents the probability of a possible POS tag.

Before applying the transformation rule, the system may have uncertainty about whether "increases" should be tagged as NNS or VBZ. A higher entropy indicates that the system is less certain about the correct tag.

After applying the transformation rule, the contextual evidence strongly supports VBZ. Therefore, the probability of VBZ becomes higher and the uncertainty decreases.

In general:

High entropy → High uncertainty → Low tagging confidence

Low entropy → Low uncertainty → High tagging confidence

Thus, Transformation-Based Tagging improves POS tagging accuracy by correcting errors using contextual rules, while entropy provides a quantitative way to measure the uncertainty of the tagging decisions before and after correction.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____